

Narrating the Endangered Species Act: A Computational Study of Narrative Policy Framework Extraction, Agenda Setting, and Policy Outcomes.

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**BOISE STATE
UNIVERSITY**

June 2026

A dissertation proposal
submitted in partial fulfillment
of the requirements for the degree of
Doctor of Philosophy in Computing
Boise State University

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Abstract

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1 Chapter 1: Introduction

Rapid biodiversity loss and growing extinction risk have made public understanding of conservation policy a critical component of environmental governance. In the United States, the Endangered Species Act (ESA) was passed by Congress in 1973 as the primary federal instrument for responding to this crisis U.S. Congress (1973). The ESA is widely regarded as one of the most powerful and controversial conservation laws, designed to prevent extinction by listing species as threatened or endangered, protecting their habitats, and directing federal agencies to avoid actions that endanger listed species Hendrickson (2005). Yet despite its scientific and legal foundations, the ESA is rarely encountered by the public through direct engagement with statutes or agency assessments. Instead, most people come to know the law through news media, which shape how conservation issues are understood, which species become visible, and what kinds of conflict define the debate Freedman *et al.* (2021).

This indirect relationship between the public and environmental policy elevates the role of media from a passive information channel to an active political actor. Media institutions translate complex scientific and legal information into accessible accounts, selectively emphasizing certain issues, actors, and interpretations while downplaying others. Through agenda-setting, the frequency and prominence of coverage shape which ESA issues and species are perceived as important, bringing some policy questions to the forefront while leaving others largely invisible McCombs & Shaw (1972). Through framing, media narratives organize these issues into coherent interpretations by defining problems, assigning causal responsibility, and highlighting particular moral evaluations Entman (1993). In ESA discourse, this often appears as competing portrayals of conservation versus economic cost, or federal protection versus local control. Through priming, repeated exposure to these frames conditions the criteria that audiences use when evaluating policies and political actors, influencing whether the ESA is judged in terms of ecological necessity, regulatory burden, or ethical obligation Iyengar & Kinder (1987). Empirical work shows that ESA coverage frequently emphasizes conflict and controversy, suggesting that media do not simply increase attention but also structure how the policy is interpreted and judged Hendrickson (2005); Hamilton *et al.* (2020b); Mutz (2015). In this sense, media do not merely report ESA debates, they actively construct the communicative environment in which those debates unfold Cox & Pezzullo (2016).

While agenda-setting, framing, and priming provide powerful tools for analyzing media influence, treating them as separate processes risks overlooking how policy communication operates in practice. In domains like the ESA, these mechanisms are intertwined, working together through broader narrative structures that link problems, actors, causes, and solutions into coherent stories. The Narrative Policy Framework (NPF) provides an integrative approach that captures this complexity by conceptualizing policy discourse as structured storytelling Jones & McBeth (2010a); Shanahan *et al.* (2018a). Within the NPF, setting directs attention (agenda-setting), characters assign roles such as heroes, villains, and victims (priming), plots establish causal relationships (framing), and policy solutions articulate normative conclusions. This integration makes the NPF particularly well suited for ESA debates, where species are often framed as victims, regulators as either protectors or overreaching authorities, and policy choices as moral judgments about conservation and economic trade-offs Bruskotter *et al.* (2018a, 2023); Nie (2004). Compared to alternative approaches such as topic modeling, which captures word co-occurrence but not causal or evaluative meaning, the NPF offers stronger analytical leverage for understanding how narratives structure policy conflict and communication Roberts *et al.*

(2014a).

Building on this framework, this dissertation examines how ESA narratives are constructed, how they circulate between media and political institutions, and whether they shape policy outcomes. Chapter 1 addresses the methodological challenge of measuring narratives at scale by developing a prompt-engineered large language model (LLM) pipeline to extract NPF elements from text and comparing it with Structural Topic Modeling (STM) as a baseline. Rather than assuming one method is superior, the chapter evaluates trade-offs in accuracy, computational efficiency, cost, and interpretability. Chapter 2 then investigates the movement of narratives across media and Congress, testing whether media lead political discourse, whether Congress drives media attention, or whether influence is reciprocal, drawing on agenda-setting and framing theories McCombs & Shaw (1972); Entman (1993); Mutz (2015). Finally, Chapter 3 links narrative discourse to policy outcomes by testing a core implication of priming: whether the moral direction embedded in media and congressional narratives predicts ESA listing decisions, including the timing and likelihood of species protection. Taken together, the dissertation argues that ESA politics cannot be understood through law or science alone, but must be analyzed as a communicative process in which narratives organize attention, interpretation, and evaluation, ultimately shaping how policy is debated and implemented Cox & Pezzullo (2016); Jones & McBeth (2010a).

2 Chapter 2: A Prompt-Engineered LLM Framework for Extracting ESA Narratives with Narrative Policy Framework-Based Coding.

2.1 Introduction

Public debates over the Endangered Species Act (ESA) unfold through dense, often fragmented bodies of text—floor speeches, committee hearings, and news coverage—that collectively construct what the law is understood to do, whom it benefits or harms, and why it should be defended or reformed Bruskotter & Shelby (2010a); Nie (2004). While the Narrative Policy Framework (NPF) provides a theoretically rich approach for analyzing these debates through structured elements such as events, causal plots, characters, and policy morals Jones & McBeth (2010a); Shanahan *et al.* (2018b), computational social science has struggled to operationalize narrative analysis at scale. Existing approaches, particularly topic models and related unsupervised techniques have become standard tools for mapping issue attention in large corpora, yet they primarily recover clusters of co-occurring words rather than the causal, relational, and evaluative structure that narrative theory treats as central Blei *et al.* (2003); Roberts *et al.* (2014b). As a result, there remains a gap between the conceptual richness of the NPF and the tools available to measure it across large collections of policy texts.

Recent advances in large language models (LLMs) offer a potential solution to this challenge by enabling the extraction of structured narrative elements directly from text. Unlike topic models, LLMs can interpret context, identify actors and actions, and generate coherent representations of events and policy reasoning Brown *et al.* (2020). However, simply applying an LLM is not sufficient. Extracting NPF elements in a consistent, replicable way requires carefully designed prompt structures that function as an algorithm for narrative extraction. In this sense, this chapter does not treat LLMs as black-box tools, but as programmable systems whose performance depends on how tasks are specified. The central methodological contribution is therefore the development of a prompt-engineered pipeline that systematically extracts ESA narrative elements (events, plots, and morals) in a way that can be reused and evaluated by other researchers. This raises an important question: can such a prompt-based algorithm provide a reliable and generalizable solution to the problem of narrative extraction in policy research?

To answer this question, the chapter evaluates the LLM-based approach against a traditional baseline: Structural Topic Modeling (STM). The goal is not to demonstrate that LLMs outperform topic models in all respects, a conclusion that would be expected given their ability to process language semantically, but to identify the conditions under which each method performs well or poorly. STM is included as a baseline because it remains widely used, computationally efficient, and highly scalable for analyzing large text corpora, even if it lacks direct access to narrative structure. This comparison therefore establishes a meaningful trade-off: while LLMs may offer richer interpretation, STM may retain advantages in terms of speed, cost, stability, and reproducibility.

This chapter evaluates competing approaches to extracting ESA narratives from large news and congressional corpora. I compare a traditional STM-based topic modeling strategy to a prompt-engineered LLM pipeline that codes NPF elements from ESA-related text. Using a

hand-checked validation sample of paragraphs drawn from newspaper articles and Congressional Record debates, I assess how often each approach correctly identifies narrative components and how much researcher intervention is required in the process. Rather than asking which method is superior, the chapter frames the comparison as a question of efficiency versus interpretive accuracy, providing a more balanced and analytically useful evaluation.

This chapter I address this through RQ1: *Can LLM prompt-based algorithms provide a reliable and generalizable solution to the problem of NPF - narrative extraction? Specifically i examine:*

- **RQ1a:** Which method better captures NPF elements (events, plots, and morals) and identifies recurring ESA narratives at scale?
- **RQ1b:** How do the methods compare in accuracy relative to human evaluation on a validation sample?
- **RQ1c:** What are the trade-offs in computational cost, interpretability, and reproducibility across media and congressional texts?

Using hand-validated comparisons across four outcome categories—LLM correct/STM incorrect, STM correct/LLM incorrect, both correct, and both incorrect—the chapter systematically evaluates performance, researcher effort, and interpretability. In doing so, it addresses a key limitation in the literature, where the NPF remains theoretically compelling but computationally difficult to implement at scale. By establishing when and how different methods can reliably measure policy narratives, this chapter provides the methodological foundation for the substantive analyses that follow.

2.2 Methods

2.2.1 Study Area

The Endangered Species Act (ESA) provides the empirical setting for this chapter because it sits at the intersection of science, politics, and moral judgment, making it one of the most narratively contested environmental policies in the United States (Nie, 2004; Bruskotter & Shelby, 2010b). Although the ESA is grounded in biological assessment and legal procedure, public understanding of the law is largely mediated through communication arenas such as news media and congressional debate, where actors repeatedly interpret what the law does, whom it benefits, and whether it represents conservation, economic burden, or federal overreach (Jones & McBeth, 2010a; Shanahan *et al.*, 2018b; Hamilton *et al.*, 2020a). This makes ESA discourse especially suitable for narrative analysis. Prior research shows that environmental policy debates are not simply collections of facts or positions, but structured stories that assign responsibility, define conflict, and embed moral evaluation (Bruskotter *et al.*, 2018b; Vucetich *et al.*, 2025). However, these narratives are embedded in large, fragmented corpora—such as congressional records and newspaper articles—that are difficult to analyze systematically at scale. Traditional computational approaches, particularly topic models, have been widely used to map issue attention in such corpora, but they primarily recover patterns of word co-occurrence rather than the causal, evaluative, and moral dimensions that define narrative structure (Blei *et al.*, 2003; Ash *et al.*, 2024a; Grimmer & Stewart, 2013a). For this reason, the ESA is not only substantively important but also methodologically demanding. It provides a setting where narrative elements—such as conflict between conservation and development, or competing claims about

responsibility—are clearly present, yet not easily captured by conventional text-as-data methods. This chapter therefore uses ESA discourse as a test case for evaluating whether newer LLM-based approaches, particularly those structured through prompt engineering, can better recover narrative meaning compared to traditional models.

2.2.2 NPF Model

This chapter is grounded in the Narrative Policy Framework (NPF), which conceptualizes policy debates as structured narratives composed of identifiable elements such as setting, characters, plot, and moral (Jones & McBeth, 2010b,d; Shanahan *et al.*, 2018c). Unlike approaches that treat policy discourse as a collection of topics or issue mentions, the NPF emphasizes how meaning is constructed through stories that define problems, assign blame, and justify solutions. The ESA provides a particularly strong application for the NPF because its debates consistently revolve around competing narratives. Species may be portrayed as victims requiring protection, regulators as either heroes or overreaching authorities, and policy interventions as either moral obligations or economic threats (Bruskotter & Shelby, 2010b; Nie, 2004; Hamilton *et al.*, 2020a). These narrative structures help explain why ESA politics remain contentious despite stable public support, as different actors mobilize competing stories to shape interpretation and evaluation (Bruskotter *et al.*, 2018b). While the NPF offers clear theoretical guidance, operationalizing its elements at scale remains a challenge. Prior work has approached this problem in several ways: conceptual coding without automation (Jones & McBeth, 2010a), structured semantic extraction (Ash *et al.*, 2024b), dynamic topic-based narrative tracking (Lange *et al.*, 2025a), and recent LLM-based pipelines for narrative identification (Schmidt *et al.*, 2025; Shahriar *et al.*, 2024a). However, these approaches either lack scalability, fail to capture full narrative structure, or rely on general-purpose LLM applications without systematic prompt design.

This chapter contributes to this gap by introducing a prompt-engineered LLM pipeline as a reproducible algorithm for extracting NPF elements from ESA discourse. Rather than treating LLMs as black-box classifiers, the approach explicitly structures prompts to operationalize narrative components—event, actors, causal story, and moral, in a way that aligns with NPF theory. This allows the study to move beyond simple application of LLMs toward a methodological contribution: demonstrating how prompt engineering can function as a theory-driven algorithm for narrative extraction. At the same time, the chapter does not assume that this approach is universally superior. Instead, it evaluates the conditions under which traditional methods, particularly structural topic modeling, may still provide advantages in efficiency, scalability, or interpretability. In doing so, the chapter reframes the comparison not as a competition between “old” and “new” methods, but as an assessment of trade-offs between computational efficiency and interpretive richness.

2.2.3 LLM Narrative Coding

The LLM component of this chapter is designed to extract narrative structure directly from text through a prompt-engineered pipeline. Drawing on advances in Large Language Models, the approach treats narrative extraction as a structured task in which the model is guided to identify specific NPF elements within each paragraph.

Rather than relying on generic prompts, this study develops a custom prompt architecture that systematically instructs the model to:

- Identify whether a paragraph is ESA-relevant
- Extract the central event
- Identify key actors
- Describe the causal story
- Assign a moral or evaluative direction

This design transforms the LLM from a general-purpose text generator into a theory-aligned coding algorithm, where each output corresponds to a defined narrative component. Importantly, the prompts are structured to preserve context and reasoning, allowing the model to incorporate the full semantic content of the text rather than relying on isolated keywords. This approach addresses a key limitation in prior computational work: while earlier methods could identify themes or entities, they often struggled to capture how those elements are connected into meaningful narratives. By contrast, the LLM pipeline is explicitly designed to recover these relationships. At the same time, LLM outputs are not treated as inherently valid. They are subject to validation against human judgments and compared with STM outputs to assess accuracy, consistency, and interpretability. This ensures that the method is evaluated critically rather than assumed to be superior.

2.2.4 Structural Topic Modeling (STM)

The Structural Topic Modeling (STM) component of this chapter serves as a baseline approach for identifying recurring patterns in ESA discourse. STM estimates latent topics based on patterns of word co-occurrence and allows researchers to examine how themes vary across documents and over time (Roberts *et al.*, 2014c; Blei *et al.*, 2003). In the context of ESA texts, STM can efficiently recover broad thematic clusters such as species listing, regulatory conflict, habitat protection, or economic impact. This makes STM well suited as a scalable first-pass method. It can process large corpora quickly, requires relatively low computational cost, and produces outputs that are transparent and reproducible. These features are particularly important given the size and heterogeneity of congressional and media datasets.

However, STM also has clear limitations for narrative analysis. While it can identify that a document discusses “regulation” or “species,” it does not capture whether the text constructs a story, assigns blame, or conveys a moral evaluation. In other words, STM operates at the level of themes rather than narrative structure. As a result, it may miss precisely the elements that the NPF treats as central. For this reason, STM is used here not as a complete solution, but as a benchmark for comparison. It allows the chapter to evaluate whether a simpler, faster, and cheaper method can approximate the insights generated by a more complex LLM-based pipeline, and under what conditions it may still be preferable.

Table 1: While previous work has made significant progress in computational narrative extraction in policy and political science, a key gap remains in the operationalization of narrative policy elements at scale. Topic models capture themes but miss narrative structure, and recent LLM approaches often lack an explicit methodological design for extracting NPF components. This chapter addresses this gap and extends this line of work by developing a prompt-engineered LLM pipeline that treats prompt design as an algorithm to extract all NPF elements simultaneously from ESA text and provides a clear basis for comparison with STM.

Study	Goal	Methods used	Reference
Jones and McBeth (2010)	Establish the Narrative Policy Framework and define the four structural elements of narrative in policy discourse.	Conceptual NPF framework; no computational extraction.	Jones & McBeth (2010c)
Grimmer and Stewart (2013)	Demonstrate the promise and limits of automated text analysis for political science research.	General text-as-data methods including topic models and supervised classification.	Grimmer & Stewart (2013b)
Roberts et al. (2014)	Model latent themes in open-ended text at scale while accounting for document-level covariates.	Structural topic modelling (STM) with metadata-informed prevalence estimation.	Roberts <i>et al.</i> (2014d)
Ash et al. (2024)	Measure interpretable narratives from plain text using structured semantic relations between entities.	RELATIO-style entity and relation extraction applied to congressional speech.	Ash <i>et al.</i> (2024b)
Shahriar et al. (2024)	Evaluate whether large language models can perform deep language understanding and context-sensitive classification tasks.	Systematic benchmarking of GPT-style models across natural language processing tasks.	Shahriar <i>et al.</i> (2024b)
Lange et al. (2025)	Track how policy narratives shift over time in large text corpora.	Dynamic topic models combined with narrative-shift detection using temporal embedding alignment.	Lange <i>et al.</i> (2025b)
Schmidt et al. (2025)	Identify economic narratives directly from large corpora without pre-specified category systems.	Integrated NLP pipeline combining entity extraction with large language model classification.	Schmidt <i>et al.</i> (2025)

Note: LLM = large language model; STM = structural topic model; NPF = Narrative Policy Framework; NLP = natural language processing. The present study differs from all prior work in applying a structured five-layer prompt to extract setting, characters, causal plot, and policy and moral simultaneously from each document, rather than modelling topics.

2.2.5 Validation Design

To assess performance, the chapter uses a hand-checked validation sample of ESA-related paragraphs drawn from both news and congressional texts. Each paragraph is manually reviewed to determine whether it contains a clear ESA narrative and, if so, what kind of narrative it represents. The computational outputs are then compared against human judgments using a four-category comparison: LLM correct/topic wrong, topic correct/LLM wrong, both correct, and both wrong. This structure allows the chapter to evaluate not only accuracy, but also the kinds of errors each method tends to make. This validation design is important because the dissertation is concerned with measurement quality, not only prediction. A method may score well overall while still missing important policy narratives, or it may recover useful narrative content but do so inconsistently across documents. The hand-checked sample therefore, provides an interpretive check on the computational methods and helps identify whether one approach is better suited to recurring ESA narratives, whether another is better at capturing structure, and which is more reproducible across corpora (Piper & Bagga, 2024; Lin & Zhang, 2025).

2.3 Results

3 Chapter 3: Agenda Setting and Framing in ESA Discourse: Who Leads the Narrative?

3.1 Introduction

Media and congressional debates about the ESA do more than register whether endangered species appear on the policy agenda; they also define what is at stake, who is responsible, and which policy solutions are reasonable McCombs & Shaw (1972); Cox & Pezzullo (2016). Agenda-setting research shows that fluctuations in coverage can shape which issues elites and the public see as important, especially for complex, low-salience problems where direct experience is rare McCombs (2014); Wanta & Ghanem (2007). Framing of narratives extends this logic by emphasizing how particular stories about problems, villains, and victims structure the space of acceptable responses Scheufele (2013); Jones & McBeth (2010a). Yet for ESA policy where public support remains stable despite polarized elite narratives Bruskotter *et al.* (2018a, 2023), a fundamental tension persists. The question of whether media drive congressional discourse by introducing new narratives, or whether Congress leads by amplifying existing frames through authoritative speech is especially consequential.

This chapter tests this tension through RQ2 : *How do narratives about the ESA evolve across media and congressional discourse, and which actor leads in shaping narrative attention and framing?*, I address two sub-questions:

- **RQ2a:** What are the 2–3 dominant ESA narratives across both corpora?
- **RQ2b:** Do media narratives precede congressional narratives or the reverse?

Building on the narrative measures developed in Chapter 1, this chapter traces how a small set of dominant ESA narratives such as the ESA listing/delisting, ESA as moral obligation circulate between news media and Congress over time Hamilton *et al.* (2020b); Boykoff & Boykoff (2007). I construct time series of LLM-coded narratives in both corpora and examine whether particular frames tend to emerge first in newspaper coverage and then appear in congressional speech, or whether the flow runs in the opposite direction. Using cross-lagged and descriptive analyses, the chapter tests agenda-setting and framing claims about whether media primarily follow congressional conflict, lead it, or co-produce the narratives through which ESA policy is contested Mutz (2015); McCombs (2014). In doing so, it treats ESA debates as a strategic case for evaluating competing accounts of who leads, who follows, and under what conditions media narratives reshape the terrain on which environmental policy conflicts unfold.

Building on the narrative measures developed in Chapter 1, this chapter traces how a set of dominant ESA narratives, including listing and delisting conflict, litigation and courts, regulatory agency action, and legislative reform battles, circulate between news media and Congress over time (Hamilton et al., 2020; Boykoff and Boykoff, 2007). I construct annual and monthly time series of LLM-coded NPF narratives in both the news articles and the Congressional Record corpus and examine whether particular narrative categories tend to emerge first in newspaper coverage and then appear in congressional speech, or whether the flow runs in the opposite direction, or whether both institutions co-produce ESA narratives through simultaneous escalation.

The primary analytical method is the Vector Autoregression (VAR) model with Granger causality testing, adapted from Barberá et al.’s (2019) cross-lagged time series approach to agenda-setting between Twitter and legislative speech. The VAR model is particularly suited to this question because it treats both media and congressional narrative series as jointly endogenous rather than designating one as the independent variable a priori, which is precisely the correct specification when the theoretical question is about the direction of influence rather than assuming it. This approach is well suited to capturing the feedback process that characterizes ESA debate, where narratives may originate in one venue, be echoed or transformed in the other, and then recirculate across the broader policy environment (Mutz, 2015). In doing so, the chapter treats ESA debates as a strategic case for evaluating competing accounts of who leads, who follows, and under what conditions media and congressional discourse reshape the terrain on which environmental policy conflicts unfold.

3.2 Methodology

3.2.1 Data

$$\text{news}_t = \alpha_1 + \beta_1 \times \text{news}_{t-1} + \beta_2 \times \text{congress}_{t-1} + \varepsilon_1 \quad (1)$$

$$\text{congress}_t = \alpha_2 + \beta_3 \times \text{congress}_{t-1} + \beta_4 \times \text{news}_{t-1} + \varepsilon_2 \quad (2)$$

3.3 Preliminary Results

4 Chapter 4: Do moral directions in narratives predict policy outcomes?

4.1 Introduction

If ESA narratives help define the stakes and meaning of endangered species protection, an important next question is whether those narratives are merely symbolic or whether they are associated with concrete policy outcomes. Priming theories suggest that sustained emphasis on particular considerations can change the criteria elites and citizens use to evaluate policies and leaders, even when aggregate attitudes or values appear stable Iyengar & Kinder (1987); Scheufele & Tewksbury (2007). In the context of the ESA, this implies that repeated narratives emphasizing economic hardship, regulatory overreach, or moral obligation might alter how quickly and under what conditions species move from scientific concern to formal listing or delisting decisions, independent of underlying biology Wymyslo (2019); Bruskotter *et al.* (2018a).

For the ESA, this creates a critical tension: **are listing decisions driven primarily by biological science and legal mandates, or do narrative environments - volume, tone, conflict - shape the timing and direction of species protections independent of ecological merit?** Wymyslo (2019); Bruskotter *et al.* (2018a).

This chapter tests narrative impact through RQ3 : *How do narratives in media and congressional discourse influence ESA listing decisions and timing?* I examine:

- **RQ3a:** Do pro-protection narratives correlate with faster listing times?
- **RQ3b:** Do conflict/anti-ESA narratives predict species listing delays or delisting attempts?
- **RQ3c:** Does narrative volume (vs silence) systematically alter listing trajectories?

This chapter examines whether narrative environments in media and congressional discourse are associated with variation in ESA listing decisions and timing. Specifically, it asks whether pro-protection narratives are linked to faster listing trajectories, whether conflict-heavy or anti-ESA narratives are associated with delays or resistance to listing, and whether narrative volume itself shapes policy movement. The core concern is whether ESA decisions are driven primarily by biological science and legal mandate, or whether the surrounding narrative context helps structure the pace and direction of species protection Jones & Rudd (2014); Manfredo *et al.* (2017, 2021).

Using ECOS/FWS species data U.S. Fish and Wildlife Service (1973–2024) matched to narrative time series, I compare **high-salience species** (polar bear, wolves-heavy narrative exposure) against **low-salience controls** (Austin blind salamander-minimal coverage) across proposal-to-listing timelines. This identification strategy tests whether narrative priming affects **decision speed**, moving narrative policy research from description to causal impact Jones & Rudd (2014); Manfredo *et al.* (2017, 2021).

4.2 Data and Methods

4.2.1 Species-Level ESA Outcomes

Species-level records maintained by the U.S. Fish and Wildlife Service and the Federal Register provide a rare opportunity to link narrative environments to observable decisions about when and whether protections are granted U.S. Fish and Wildlife Service (1973–2024).

To connect narratives to policy outcomes, I use species-level records from the U.S. Fish and Wildlife Service and related Federal Register notices. These records provide the key outcome data for the chapter, including proposal dates, final listing dates, delisting actions, and other major ESA decisions. Each species is matched to the surrounding narrative environment based on species names and policy references found in the text corpus. This creates a species-level dataset in which the dependent variable is the timing or direction of ESA action, and the explanatory variables are the narratives associated with that species.

4.2.2 Survival Analysis Framework

The main empirical strategy uses survival analysis to estimate whether narrative environments are associated with the timing of ESA listing decisions. In this framework, the event of interest is the first formal ESA action, such as listing or delisting, and the clock begins when a species enters the relevant policy process. Species that do not receive a final decision by the end of the observation window are treated as censored cases. This approach is appropriate because it preserves the timing information in the data and allows for direct testing of whether narrative exposure accelerates or delays policy outcomes.

4.3 Preliminary Results

QUESTIONS or Areas for Your Thoughts:

- 1. For Chapter 3, I am trying to decide whether survival analysis is the best fit for examining how narratives relate to ESA listing timing and decisions, or whether I should incorporate machine learning methods to strengthen prediction. Do you think one approach is more appropriate for the chapter's argument and research design?**

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